

Table 1. False-premise detection across models and datasets

Method	Qwen2.5 7B				GPT-6 Luna				Gemma 4 12B				Qwen3.8 27B OFF				Qwen3.8 27B ON			
	Cancer-Myth		CREPE		Cancer-Myth		CREPE		Cancer-Myth		CREPE		Cancer-Myth		CREPE		Cancer-Myth		CREPE	
	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓
TF-IDF text	Shared across models — Cancer-Myth: TPR ↑ 86.1% (H), FPR ↓ 51.0% (H); CREPE: TPR ↑ —, FPR ↓ —																			
Direct gate	7.0 ^H	10.7 ^H	—	—	—	—	—	—	40.5 ^H	16.1 ^H	—	—	26.1 ^H	16.1 ^H	—	—	44.1 ^H	21.5 ^H	—	—
Premise-review CoT gate	29.5 ^H	16.1 ^H	—	—	—	—	—	—	51.3 ^H	32.9 ^H	—	—	57.5 ^H	35.6 ^H	—	—	61.4 ^H	38.3 ^H	—	—
Probe gate (hidden states)	75.0 ^H	47.7 ^H	—	—	n/a	n/a	n/a	n/a	—	—	—	—	—	—	—	—	—	—	—	—
GEPA	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Ours	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Stored results; —: not measured/comparable; n/a: outside the defined implementation. Percentages, one run.
H: historical full-corpus Cancer-Myth; valid denominators and protocol differences in ledger.
TPR ↑: FPQ detection; FPR ↓: false positives on normal questions.
Probes require hidden states; no native Luna probe. TF-IDF is model-independent.
Default crossfit probe/TF-IDF decisions; these are not matched-FPR thresholds.
Ours denotes the proposed method; its implementation is not fixed.